

CrOSSD

Towards a Critical Open-Source Software Database

Sebastian Neumaier, Tobias Dam

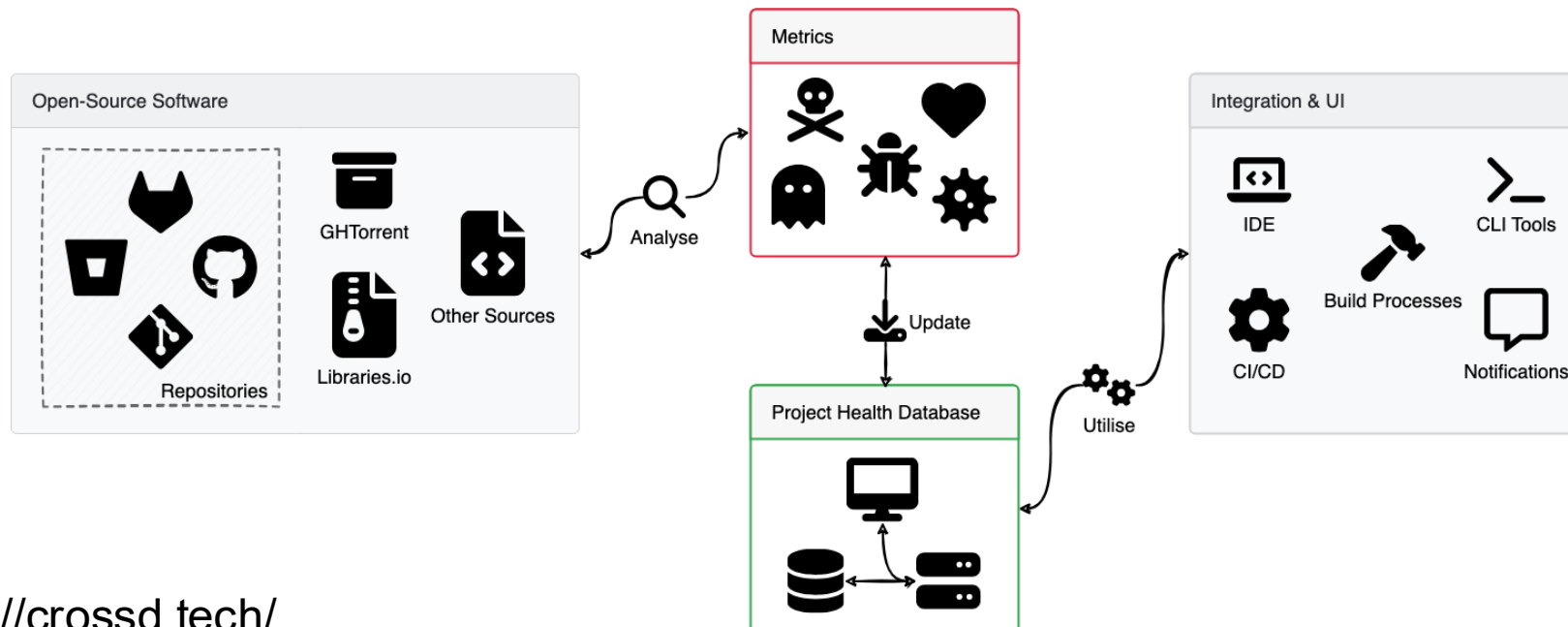
University of Applied Sciences St. Pölten, Austria

CHAOSScon-2026-EU, 29 January 2026



Assessment of OSS Health

- Transparent, evidence-based assessment of OSS project health
 - **Quantitatively & automated**: collection of meta information
 - **Qualitatively**: sustainability/funding, activity indicators, security policies, etc.



Metrics Methodology

- **Define metric concept**

- Literature-driven approach through review of research on OSS health & criticality
- Strongly aligned/influenced by CHAOSS catalog: e.g. community activity, development dynamics, security exposure, and governance/compliance

- **Identify measurable signals**

- Github APIs, metadata on dependencies, vulnerabilities, etc.

- **Crawling, archiving, and metric computation**

Challenges

- **Scalability**

- Repositories rate limits
- Limited to selected projects due to storage constraints

- **Access to relevant data**

- Missing information (e.g. funding, organisations behind projects)
- No structured/standardized endpoints for key metadata

- **Hard-to-measure aspects**

- E.g. documentation quality
- Community health, e.g. in terms of collaboration practices

Interactive Demo

- Web UI
- Use of API
 - JupyterLite Demo
 - <http://tiny.cc/h05y001>



Stakeholder Interviews

22 semistructured interviews

- Participants from across the OSS ecosystem (mostly software developers; a few from adjacent roles)

Topics:

- Definitions and understandings of OSS quality and project health
- Conceptualisations of *criticality*
- Needs, expectations, feature requests w.r.t. CrOSSD

Stakeholder Interviews

Current state:

- Transcription (using noScribe) ongoing
- Thereafter, inductive open coding and analysis

Outreach:

- Calls via networks & social media
- Intentional effort to reach non-male contributors

Overall picture:

- Good diversity overall
- Underrepresented groups are primarily older people and (still) non-males

Stakeholder Interviews – First Results

- **OSS quality:**
 - Broad agreement on common factors (popularity/usage, documentation, contributors, activity)
 - Plus a few interesting outliers
- **Criticality:**
 - Few had a predefined concept
 - Recurring criteria: (1) how widely used, (2) severity of impact in case of failure/breakage/disappearance, (3) replaceability, (4) security relevance
- **CrOSSD:**
 - Opinions ranged from “irrelevant/not useful to me” to “highly important”
 - Provided diverse input shaping future development

Next Steps for CrOSSD

- Finalizing interview study
 - Analysis and publication
- Integration of other code repositories
 - Gitlab, etc.
- Work on "AI-based" metrics
 - E.g. analysis of tone in issues using LLMs
 - Utilize RAG to counteract context problems
 - Challenge of potentially large projects

Contact

Sebastian Neumaier

sebastian.neumaier@ustp.at

Tobias Dam

tobias.dam@ustp.at

<https://crossd.tech/>

<https://health.crossd.tech/>

<https://github.com/FH-CrOSSD>

